

Adina Nadeem

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Summary

MSc Bioinformatics student at Freie Universität Berlin and Bioinformatics Student Assistant at the Max Planck Institute for Infection Biology. I am interested in machine learning for biology, particularly how learned representations capture biological structure across protein sequences and cellular states, and how those representations behave across different contexts and perturbations. My current work spans protein language models, computational genomics and reproducible bioinformatics research.

Education

M.Sc. Bioinformatics

Oct 2024 – Present

Freie Universität Berlin, Germany

Specialisation in Data Science. Expected completion: 2027.

B.S. Bioinformatics

2019 – 2023

COMSATS University Islamabad, Pakistan

Technical Skills

Programming: Python, R, Bash; working knowledge of SQL.

Machine learning & data analysis: supervised learning, feature engineering, model evaluation and interpretation, dimensionality reduction, clustering; protein language models (ESM2); basic PyTorch for tensor operations and model inference.

Bioinformatics & genomics: bulk and single-cell RNA-seq, amplicon sequencing, multi-omics, protein sequence and structure analysis, molecular docking, genome assembly and annotation; DESeq2, Seurat, Salmon, SAMtools, Biopython.

Scientific computing: Nextflow, Snakemake, Docker, Apptainer, Conda, Git, Linux, Slurm/HPC; AWS EC2 and S3.

Certification: AWS Certified Cloud Practitioner.

Experience

Bioinformatics Student Assistant

Sep 2025 – Present

Max Planck Institute for Infection Biology, Berlin | Portugal Group

- Develop computational analyses and research software for longitudinal *P. falciparum* studies of persistence and within-host dynamics in Mali, working closely with PhD researcher [Manuela Carrasquilla](#).
- Built desktop and Slurm/HPC workflows for amplicon sequencing data, including containerised execution, resumable runs, sample-sheet preparation and quality-control reporting.
- Integrate genotyping outputs, participant metadata and replicate checks for downstream analyses of new versus persistent infections, infection complexity and parasite relatedness.

Student Researcher – BUA StuROPx

Oct 2025 – Jan 2026

Charité – Universitätsmedizin Berlin

Topic: Machine learning for HPV16 T-cell epitope prioritisation

- Carried out the computational analysis for HLA-specific prioritisation of HPV16 T-cell epitopes using IEDB data, BLOSUM62 encodings, and physicochemical and immunological features.
- Trained and evaluated LightGBM and random forest models to predict immunogenicity scores and rank candidate epitopes; co-authored the resulting research poster.

Bioinformatics Junior Scientist

Oct – Dec 2023

Bio-code

- Developed Python, Bash and R workflows through projects in whole-genome sequencing, metagenomics and transcriptomic data analysis.

Experience (continued)

Undergraduate Researcher

Aug 2022 – Feb 2023

Department of Biosciences, COMSATS University Islamabad

- Developed a reproducible Nextflow workflow to identify polyadenylated reads from RNA-seq data, integrating adapter trimming, strand inference, poly(A/T) detection and read alignment.
- Benchmarked adapter-trimming and strand-detection tools using simulated and public RNA-seq datasets to select components for the workflow.
- Automated data retrieval and processing with Python and packaged the workflow and dependencies in Docker for reproducible execution.

Selected Research Projects and Software

Protein language model representations and sequence landscapes

Ongoing

ESM2, Python, evolutionary sequence search, ESMFold/AlphaFold

Started as a deep-learning module project on a topic proposed by [Luca Ferretti](#), [Vitaly Belik](#) and [Alisa Sergeeva](#); now continuing as a collaborative research project being developed towards a manuscript.

- Develop and extend a computational sequence-search framework using pretrained ESM2 representations to explore protein sequence landscapes.
- Implemented adaptive mutation refinement and analyse how sequence context and epistatic effects influence search trajectories across different proteins.
- Compare recovery in representation and sequence space with ESMFold/AlphaFold-based structural comparisons to test whether progress according to the model also corresponds to recovery of protein structure.

malaria-amplicon-nf

Desktop | HPC

Portugal Group | Amplicon analysis software for desktop and HPC use

- Built a desktop application for analysing data from the six-locus SIMPLseq amplicon panel, with FASTQ scanning, sample-sheet preparation, Nextflow execution and quality-control reports.
- Added Slurm/HPC execution with Apptainer packaging, resource configuration, resumable runs and automated collection of output tables and summaries.
- Integrated DINEMITES and Dcifer using allele tables, participant metadata and replicate checks to analyse new versus persistent infections, infection complexity and parasite relatedness.

Single-cell var-gene expression analysis

Source figure

Portugal Group | Python, R, single-cell RNA-seq, clustering, data visualisation

- Built a reusable workflow to reproduce the analyses and visualisations behind Figure 6 of the Portugal group's published study of var-gene expression in asymptomatic and clinical malaria infections.
- Combined expression matrices and cell metadata to generate clustering, annotated heatmaps and domain-coverage plots, with configurable inputs and exported plot data for reuse.

Multi-omics machine learning for breast cancer subtypes

GitHub

TCGA-BRCA, RNA-seq, DNA methylation, machine learning, SHAP

- Compared molecular feature sets and supervised classifiers for Basal, Luminal A and Luminal B subtype prediction using TCGA-BRCA expression, methylation and clinical data with holdout evaluation.
- Used SHAP and pathway enrichment to interpret predictive features; examined subtype differences in epigenetic age acceleration and estimated immune-cell fractions.

Professional Development

Google Summer of Code Proposal – Wellcome Sanger Institute

2024

- Developed a proposal to modernise a pathogen-focused mapping and variant-calling pipeline in Nextflow, including modular workflow design, containerisation and pseudosequence/phylogenetic outputs.
- Refined the technical design with Sanger mentors; the proposal was forwarded by the mentors for GSoC consideration but was not selected by Google in the final round.